Jayant Unde

Curriculum Vitae

Nagoya University,
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Education

Oct 2023–May 2025 **PhD, Micro-Nano Mechnical Science and Engineering**, *Nagoya University*, Nagoya, Japan.
Thesis title: Robot-Assisted Dressing Systems with Transforming Linkage Mechanisms for Enhancing Elderly Independence in Toileting, *Advisor:* Prof. Yasuhisa Hasegawa, *GPA:* **4.23/4.3.**

Oct 2021–Sept 2023 **Master of Engineering, Mechanical Systems Engineering**, *Nagoya University*, Nagoya, Japan.
Thesis title: Development of Novel Linear Actuator using Articulated Lead Screw Mechanism
Advisor: Prof. Yasuhisa Hasegawa, *GPA:* **4.11/4.3.**

June 2014–May 2018 **Bachelor of Engineering, Mechanical Engineering**, *COEP Tech. University*, *formerly* College of Engineering, Pune (COEP), India.
Advisor: Prof. S. S. Ohol, *CGPA:* **8.51/10** – Graduated with Honors

Research Experience

May 2025 – Present **Postdoctoral Researcher**, *Nagoya University*, Japan.
- Leading and mentoring graduate students within the JST Moonshot project, overseeing mechanism design, prototyping, and experimental validation.
- Researcher in the international DEMETER project, collaborating with teams in Japan, France, and Germany to develop next-generation agricultural robots for autonomous vineyard pruning.
- Designing compliant end-effectors for deformable object manipulation, with applications to robot-assisted dressing and safety-critical human interaction.

Oct 2023 – May 2025 **PhD Researcher**, *Nagoya University*, Japan.
Part of one of Japan's largest national initiatives: JST Moonshot R&D Program
- Led Moonshot research at Nagoya University in collaboration with multiple Japanese universities, advancing next-generation assistive robots for Activities of Daily Living.
- Invented a high-extension linkage-based linear actuator and developed passively adaptive grippers integrated into robotic dressing-assistance systems for safe manipulation of deformable fabrics.
- Disseminated research through peer-reviewed journal publications and presentations at major international robotics conferences, including **ICRA** and **IROS**.

Jul 2024 – Nov 2024 **Visiting Research Intern**, *Polytechnique Montréal*, Canada.
Advisor: Prof. Carl-Éric Aubin, Prof. David Mélançon
- Conducted interdisciplinary research on shape-morphing metamaterials for orthopedic applications.
- Developed a configurable orthopedic brace for scoliosis using 3D-printed TPU inflatables.
- **Outcome:** Patent filed for the proposed orthopedic brace concept.

Nov 2020 – Oct 2021 **Research Student**, *Nagoya University*, Japan.
Advisor: Prof. Yasuhisa Hasegawa
- Conducted preliminary research on robotic actuators under MEXT Scholarship.

Industrial Experience

Dec 2022 – Apr 2023 **Digital Transformation Intern**, *Mitsubishi Fuso Truck and Bus Corp*, Kawasaki, Japan.
- Assisted in developing a digital platform to enhance customer service by reducing vehicle downtime.
- Led user acceptance testing discussion with internal & external stakeholders.

Jul 2018 – Oct 2020 **Mechanical Design Engineer**, *Aker Solutions*, Pune, India.
- Designed subsea XMT equipment for large deep-water oil & gas projects.
- Received **'Significant Contribution Award'** for resolving critical engineering bottlenecks.

Selected Fellowships, Prizes, and Awards

2025 Young Researcher Award, IEEE RAS Japan Joint Chapter (IEEE IROS 2025).

2025 Young Researcher Encouragement Award, IEEE Nagoya Section.

2023 Best Paper Award, IEEE CBS 2024, Nagoya

2023 Best Paper Award, IEEE RO-MAN 2023, Busan.

2021–2025 MEXT Scholarship, awarded by the Japanese Government for Higher Education in Japan.

2018 3rd Prize, out of 1088 other teams at DRDO Robotics and Unmanned Systems exposition.

2017 Represented Team India at the ABU Robocon International Competition, Tokyo, Japan.

2017 National Winner, Robocon India (National Robotics Competition).

2016 National 2nd Place, Robocon India.

Selected Publications

Journal Articles

2025 **J. Unde,**, Y. Ito, J. Colan, and Y. Hasegawa, "Kinematic Analysis and Experimental Verification of Transforming Planar Linkage Mechanism," *ROBOMECH Journal*, vol. 12, no. 3, 2025.

2024 **J. Unde,** J. Colan and Y. Hasegawa, "Novel Articulated Lead Screw Linear Actuator Enabled by Transforming Linkage Mechanism," in *IEEE Robotics and Automation Letters*, vol. 10, no. 2, pp. 1369-1376, 2025. (Presented at IEEE **IROS 2025**.)

2024 **J. Unde,** J. Colan and Y. Hasegawa, "Design, Modeling, and Experimental Verification of Passively Adaptable Roller Gripper for Separating Stacked Fabric," in *IEEE Robotics and Automation Letters*, vol. 9, no. 11, pp. 9335-9342, Nov. 2024. (Presented at IEEE **ICRA 2025**.)

2024 **J. Unde,** T. Inden, Y. Wakayama, J. Colan, Y. Zhu, T. Aoyama, and Y. Hasegawa, "Development of a compliant gripper for safe robot-assisted trouser dressing-undressing," *Advanced Robotics*, vol. 38, no. 19–20, pp. 1424–1440, 2024.

2024 S. Sugiura, **J. Unde,** Y. Zhu and Y. Hasegawa, "Variable Grounding Flexible Limb Tracking Center of Gravity for Sit-to-Stand Transfer Assistance," in IEEE Robotics and Automation Letters, vol. 9, no. 1, pp. 175-182, Jan. 2024.(Presented by the applicant at IEEE **ICRA 2024**).

In Conference Proceedings

2025 **J. Unde,** T. Urata, S. Kamei, Y. Ito, R. Kihara, J. Colan, and Y. Hasegawa, "Design and Preliminary Evaluation of a Walker-Mounted Robotic System for Elderly Toilet Dressing Assistance," 2025 34th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), Eindhoven, Netherlands, 2025, pp. 73-78.

2024 Y. Ito, **J. Unde**, S. Sugiura, T. Urata and Y. Hasegawa, "Design and Strength Verification of a Compact VGFL for Sit-to-Stand Assistance Robot*," 2024 IEEE International Conference on Cyborg and Bionic Systems (CBS), Nagoya, Japan, 2024, pp. 273-278. **Best Paper Award**

2023 **J. Unde,** J. Colan, Y. Zhu, T. Aoyama and Y. Hasegawa, "Single Actuator Tendon Driven Two Finger Linkage Gripper with Strong Pinch and Adaptable Cylindrical Grasp," 2023 32nd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), Busan, Korea, Republic of, 2023, pp. 119-125. **Best Paper Award**

2021 **J. Unde,** J. Colan and Y. Hasegawa, "Bidirectional Tether Less Soft Actuator with Expeditious Position Control Mechanism," The 32nd 2021 International Symposium on Micro-NanoMechatronics and Human Science, Nagoya, Japan, 2021, pp. 1-6.

Patents

2025 **J. Unde**, T. Aoyama, and Y. Hasegawa, *"Link mechanisms, link systems, linear actuators and body support devices"* (in Japanese), Patent No. 2024-035514. Status: Under review. Filed: September 5, 2022.

2025 **J. Unde**, and Y. Hasegawa, *"Link mechanism, link system and actuator"* (in Japanese), Patent No. 2025-203957. Status: Filed. Filed: November 26, 2025.

Skills

Programming	Python, C++, MATLAB Simulink, LaTeX
Robotics	ROS, CoppeliaSim, Kinematics, Dynamics
Engineering	SolidWorks (Certified Solidworks Professional), ANSYS (Static/Topology), Rapid Prototyping
Languages	English (Professional), Japanese (JLPT N2 - Advanced), Marathi/Hindi (Native)

Service to the Scientific Community

2022–Present Reviewer for:
- Springer Nature– ROBOMECH Journal
- IEEE Robotics and Automation Letters (RA-L)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)

2025 Invited Live Demonstration at Swiss Pavilion, Osaka World Expo 2025, for the Symposium on *AI & Robotics in Assistive Technology: Swiss-Japanese Perspectives* (Swiss Embassy News)

2025 Session Chair for Mechanism Design and Optimization session at the IEEE CBS 2025 Conference, Beijing, China.